Nepal's Role in Zebrafish Science

Where Nepal sits in the natural range of Danio rerio, and what is known and not yet known

Nepal is part of the home range of the zebrafish. This paper brings together what is known about the species in Nepal's rivers, the evidence from genetic and fish surveys, and the gaps that local research can fill.

NZINC Nepal Zebrafish in Nature Center

Bharatpur-7, Chitwan, Nepal

1 Nepal in the natural range

Nepal is one of the countries where the zebrafish occurs naturally, and its rivers lie in the heart of the species' home region.

The zebrafish is native to freshwater habitats in India, Pakistan, Bangladesh, Nepal and Bhutan, with populations concentrated in the Ganges and Brahmaputra basins. The species was first described from the Kosi River, which drains eastern Nepal before joining the Ganges in India. Many of Nepal's rivers, including the Narayani, Koshi and Karnali systems, are tributaries of the Ganges and belong to the same river basin as the original type locality.

FACT	DETAIL
Native range	India, Pakistan, Bangladesh, Nepal, Bhutan
Main river basins	Ganges and Brahmaputra
Typical habitat	Shallow, clear, slow or still water: streams, ponds, ditches, rice paddies
Elevation recorded	Up to 1,795 m above sea level
Survey water temperatures	About 16.5 to 34 °C
Wild adult size	1.8 to 3.7 cm
Local names in Chitwan	*Khida* (*Danio rerio*) and *Chaiyapothi* (*Danio devario*)

Sources: Wikipedia, Zebrafish; Dhital, Fisheries Development Centre, Hetauda (FAO).

2 Evidence from Nepal

The population genomic study by Whiteley and colleagues (*Molecular Ecology*, 2011) included wild populations from Nepal, India and Bangladesh. Of the 13 wild populations sampled, the study found five genetically distinct groups matching separate geographic regions. The result shows that wild zebrafish from different parts of the range differ from each other, and Nepal's populations were part of the sample.

3 The Narayani survey

A fish survey of the Narayani River system in Chitwan, carried out by the Fisheries Development Centre in Hetauda, gives one of the clearest local pictures. The Narayani runs about 332 km through Nepal, fed mainly by the Kaligandaki, and the survey sampled it together with the Trishuli, Rapti and several smaller streams and tributaries.

SPECIES	ORDERS	FAMILIES	CYPRINID SPECIES
69	9	19	34

Table 1. Fish collected in the Narayani River system, Chitwan. Source: Dhital, Fisheries Development Centre, Hetauda.

Both *Danio rerio* and *Danio devario* were described as common there, mostly found during the low-water season, from October to January. Other small, abundant fishes in the same waters included *Puntius* species, *Rasbora daniconius* and *Esomus danricus*. Earlier surveys had reported between 23 and 111 species, depending on the elevation range and the method used. This variation shows how much the picture depends on survey effort.

4 Pressures

The same survey identified the main threats to fish in the Narayani system: unregulated fishing, dynamiting and poisoning, taking of broodstock during spawning, small-mesh nets, deforestation and silting, and water pollution from industrial waste and agricultural chemicals. Seasonal changes in water level and winter drying add natural stress, especially in small streams.

5 Why Nepal matters to zebrafish science

- **Natural variation.** Wild populations carry genetic variation that is missing from laboratory strains. Nepal's populations add to the sample of the species' range.
- **Habitat links to agriculture.** Zebrafish live in the shallow margins and paddies that overlap with farmland, which makes them relevant to studies of pesticide use, water management and food security.
- **Opportunity for baseline data.** In the sources reviewed for this paper, Nepal-specific data on zebrafish distribution and population genetics are limited. Systematic surveys across the Terai, the mid-hills and different seasons would fill a clear gap.
- **Research partnerships.** The University of Oregon, where laboratory zebrafish work began, and Duke University are among the institutions with which NZINC has collaborative links, together with Agriculture and Forestry University in Nepal.

6 The work of NZINC

Nepal Zebrafish in Nature Center (NZINC) is a non-profit research organisation based in Bharatpur, Chitwan. Its work on zebrafish and aquatic biodiversity includes studying zebrafish populations in their natural habitats through morphometric characterisation, habitat assessment and water quality analysis, and applying molecular methods to characterise genetic and phenotypic diversity. NZINC has signed a memorandum of understanding with Duke University to advance research on native zebrafish, aquatic biodiversity and river ecosystem conservation in Nepal, and an agreement with Agriculture and Forestry University to promote joint research and academic exchange.

Next steps for Nepal

- Map confirmed zebrafish sites by river basin, elevation and season.
- Collect samples for genetic analysis and compare them with populations elsewhere in the range.
- Record water quality and land use at each site.
- Share results with universities, local government and communities along the rivers.

Sources

Whiteley A.R. et al. (2011). Population genomics of wild and laboratory zebrafish (*Danio rerio*). *Molecular Ecology* 20(20): 4259–4276.

Dhital R.R., Fisheries Development Centre, Hetauda, Nepal. Fish survey of the Narayani River system, Chitwan (FAO, fao.org/4/y3994e/y3994e0e.htm).

Zebrafish. Wikipedia (en.wikipedia.org/wiki/Zebrafish); Animal Diversity Web, *Danio rerio* (animaldiversity.org/accounts/Danio_rerio).

NZINC. Website information on research focus, the Duke University MoU and the agreement with Agriculture and Forestry University.